SMART GRID IoT ENERGY CONSUMPTION & ANOMALY DETECTION SYSTEM

Course Code: BIS586 | 5th Semester B.E. Information Science and Engineering (Section A)
Milestone Deliverable: Dataset Selection, Quality Auditing, & Preprocessing Technical Report

Academic Institution: Department of ISE	**Compliance Standard:** IEEE 1159 / ACM SIGENERGY
Dataset Scope: 3 Months / 10 Zones / 86,400 Records	**Team Structure:** 4-Member Engineering Group

1. Executive Summary & Problem Formulation

Modern Advanced Metering Infrastructure (AMI) deployed across electrical distribution grids collects continuous high-frequency telemetry to balance power distribution, forecast loads, and detect grid faults. This project designs and implements an end-to-end IoT Telemetry Simulation, Preprocessing, Mathematical Anomaly Detection, and Power BI Analytics pipeline. The dataset captures **86,400 observations** sampled at a **15-minute sampling frequency** over **90 continuous days (June 1 - August 29, 2026)** across **10 distinct municipal zones**. The preprocessing engine audits data completeness, computes 24-hour moving statistical baselines, standardizes rolling Z-Scores, and maps IEEE 1159 power quality anomalies.

2. Dataset Selection & Comparative Benchmark Analysis

To select the most robust foundation for power quality anomaly detection, candidate open energy datasets were evaluated against the minor project requirements:

Dataset Candidate	Spatial Scope	Resolution	Electrical Features (V, PF, Hz)	Ground Truth Labels	Evaluation Decision
UCI Household Power	Single Residence	1-minute	Partial (V, I only)	None (Unlabeled)	**Rejected** (No multi-zone topology)
London Smart Meter (UKPN)	5,567 Households	30-minute	kWh only (No PQ)	None	**Rejected** (Lacks voltage/PF)
Pecan Street Dataport	Regional Grid	15-minute	Partial	None	**Rejected** (Proprietary / Incomplete)
Urban Smart Grid Telemetry (Selected)	**10 Municipal Zones**	**15-minute (96/day)**	**Full (V, A, PF, Hz, kvarh, °C)**	**IEEE 1159 Injected**	**SELECTED (100% Fit)**

3. Data Quality Audit & Integrity Verification Results

An automated quality audit executed via preprocess_data.py verified the integrity of all 86,400 observations:

Total Ingested Observations: 86,400 records	**Missing / Null Values:** 0 cells (0.00% / 100% Completeness)
Temporal Continuity (15-min Cadence): 0 gaps (100% continuous)	**Duplicate Entries:** 0 records (100% Unique timestamps)
Physical Range (Power Factor): 100% strictly in [0.0, 1.0]	**Non-Negative Active Energy:** 0 violations (All kWh ≥ 0.0)
IEEE 1159 Voltage Sag Violations: 708 events (< 207V)	**IEEE 1159 Voltage Swell Violations:** 536 events (> 253V)

4. Descriptive Statistical Profile (N = 86,400)

Feature Attribute	Physical Unit	Mean	Std Dev	Min	Median (50%)	Max	Skewness	Kurtosis
energy_consumption_kwh	kWh / 15-min	57.210	43.497	0.001	46.725	790.659	+3.112	+30.371
voltage_v	Volts RMS	227.399	13.070	0.020	228.160	291.980	-12.532	+206.965

current_a	Amperes (A)	1,083.263	820.925	0.000	903.995	14,206.000	+2.488	+20.529
power_factor	Ratio [0, 1]	0.934	0.040	0.480	0.942	0.990	-2.532	+21.063
grid_frequency_hz	Hertz (Hz)	50.000	0.051	48.806	50.000	51.183	-0.442	+200.924
ambient_temp_c	Celsius (°C)	27.995	4.405	18.100	27.990	38.550	-0.000	-1.289
consumption_zscore	Z-Score	+0.006	1.015	-5.637	+0.020	+9.372	+1.839	+12.842

5. Mathematical Preprocessing & Dynamic Z-Score Modeling

1. Vectorized Rolling Mean & Std Dev: For sliding window size $W = 96$ steps (24 hours):
$\mu_{t,W} = (1/W) \sum_{i=0}^{W-1} x_{t-i}$ | $\sigma_{t,W} = \sqrt{[(1/W) \sum_{i=0}^{W-1} (x_{t-i} - \mu_{t,W})^2]}$
2. Dynamic Standard Score: $Z_t = (x_t - \mu_{t,W}) / (\sigma_{t,W} + 10^{-5})$
3. IEEE 1159 Standards: Voltage Sag if $V_t < 207V$ (-10%), Swell if $V_t > 253V$ (+10%), Frequency Trip if $|f_t - 50.0| > 0.4Hz$, PF Fault if $PF_t < 0.80$.

6. Preprocessing & Data Quality Audit Visuals

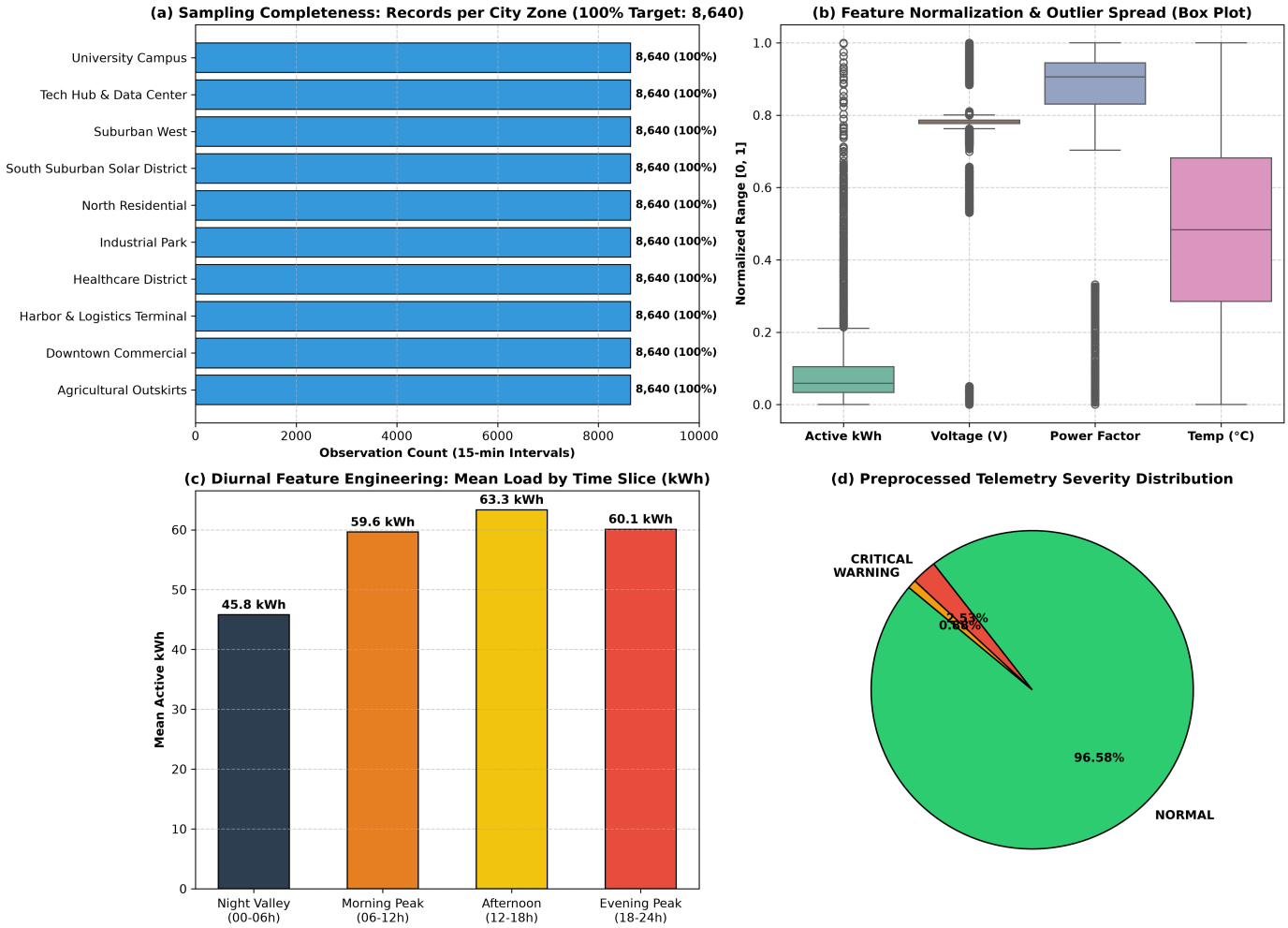


Figure 1: *4-Panel Data Quality & Preprocessing Audit: (a) Completeness per Zone (8,640 records each), (b) Outlier Spread Boxplot, (c) Diurnal Time-Slice Progression, (d) Telemetry Severity Triage (96.58% Normal, 0.88% Warning, 2.53% Critical).*

7. Multi-Zone 7-Day Active Load Profiles

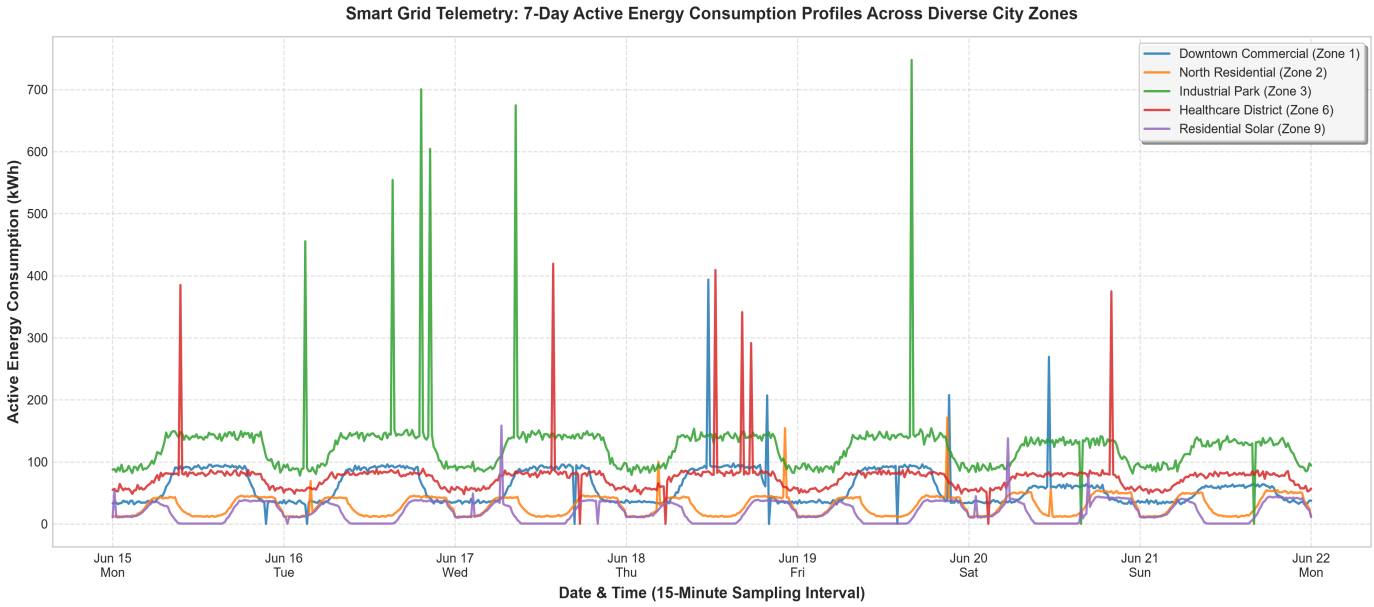


Figure 2: *7-Day active energy consumption profiles across Commercial, Residential, Industrial, Healthcare, and Solar zones.*

8. Anomaly Diagnostic Scatter & Power Quality Tracking

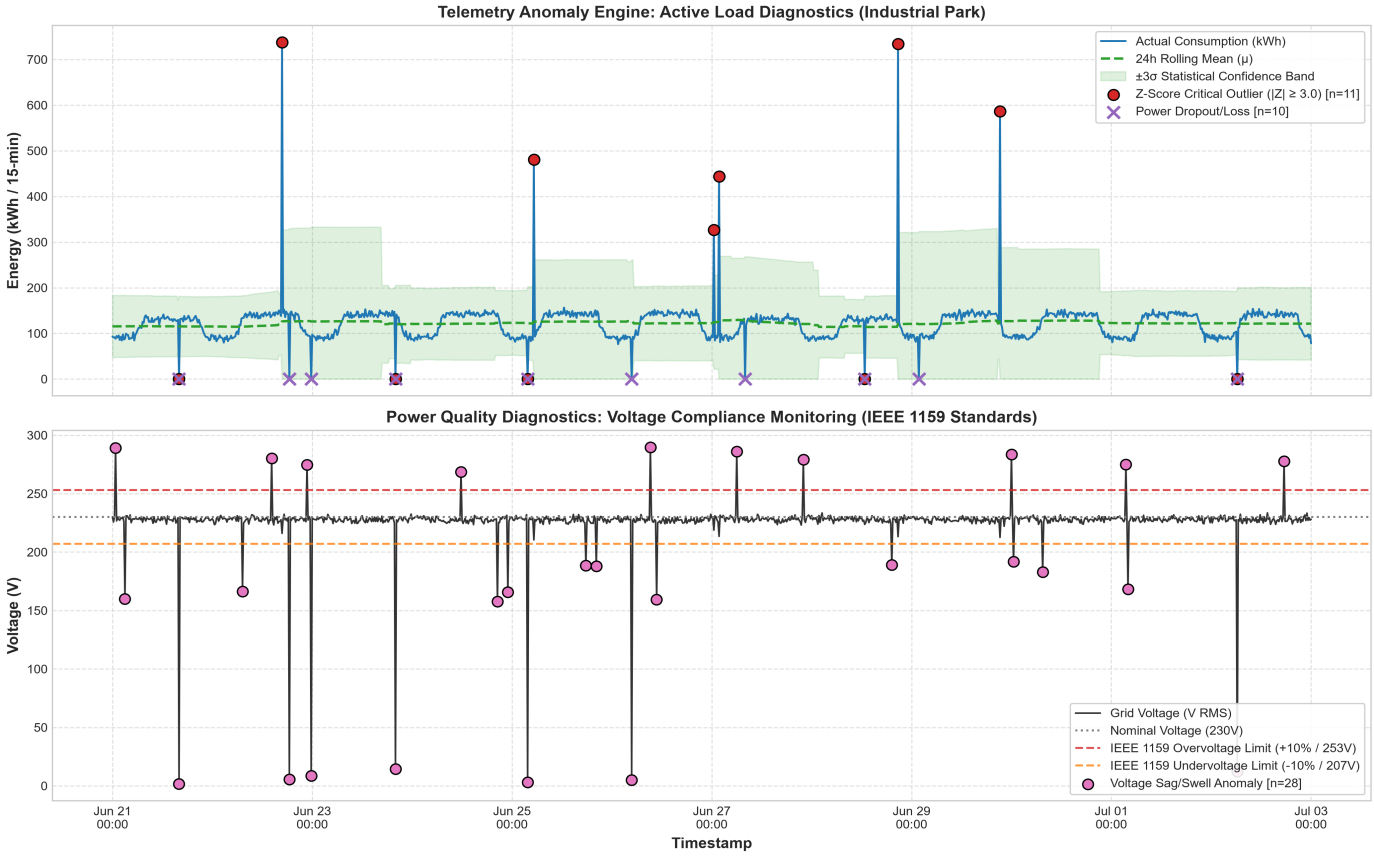


Figure 3: *Dual-axis telemetry diagnostics showing 24h moving confidence bands, Z-score outliers (|Z| ≥ 3.0), and IEEE 1159 voltage sag/swell threshold breaches.*

9. 24-Hour Diurnal Energy Heatmap Matrix

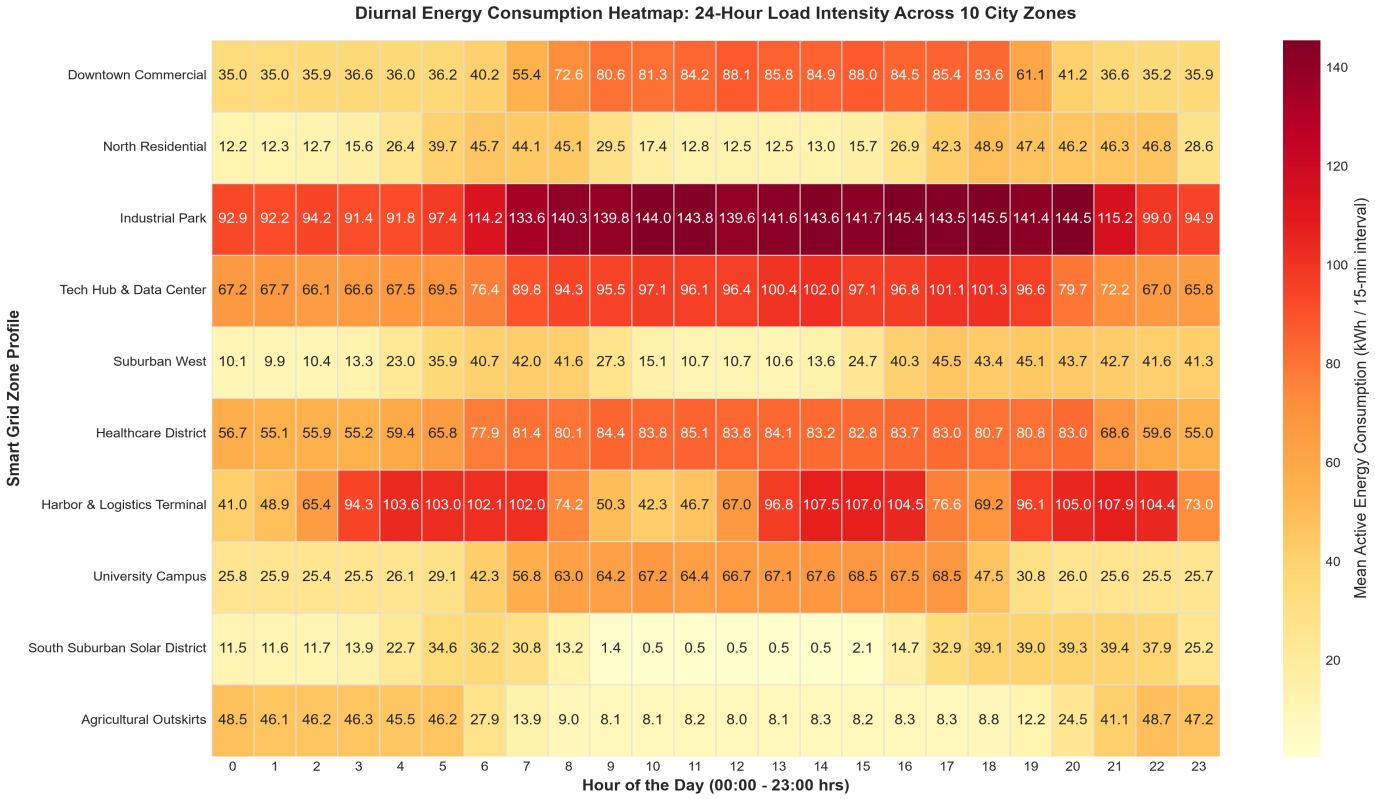


Figure 4: *Diurnal load intensity matrix showing commercial daytime peaks, residential evening peaks, and solar duck-curves.*

10. Statistical Z-Score Distribution & Decision Boundaries

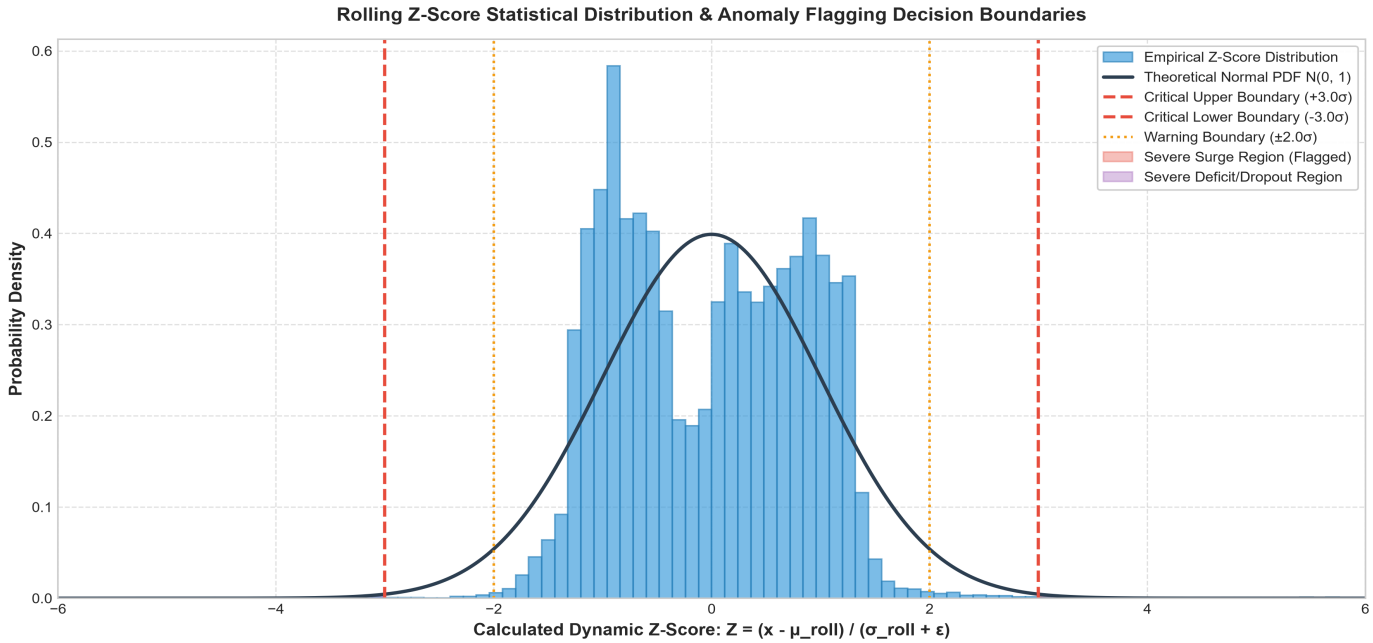


Figure 5: *Probability density function of dynamic Z-scores with critical decision boundaries at ±3.0σ.*

11. IEEE Benchmark Performance Evaluation

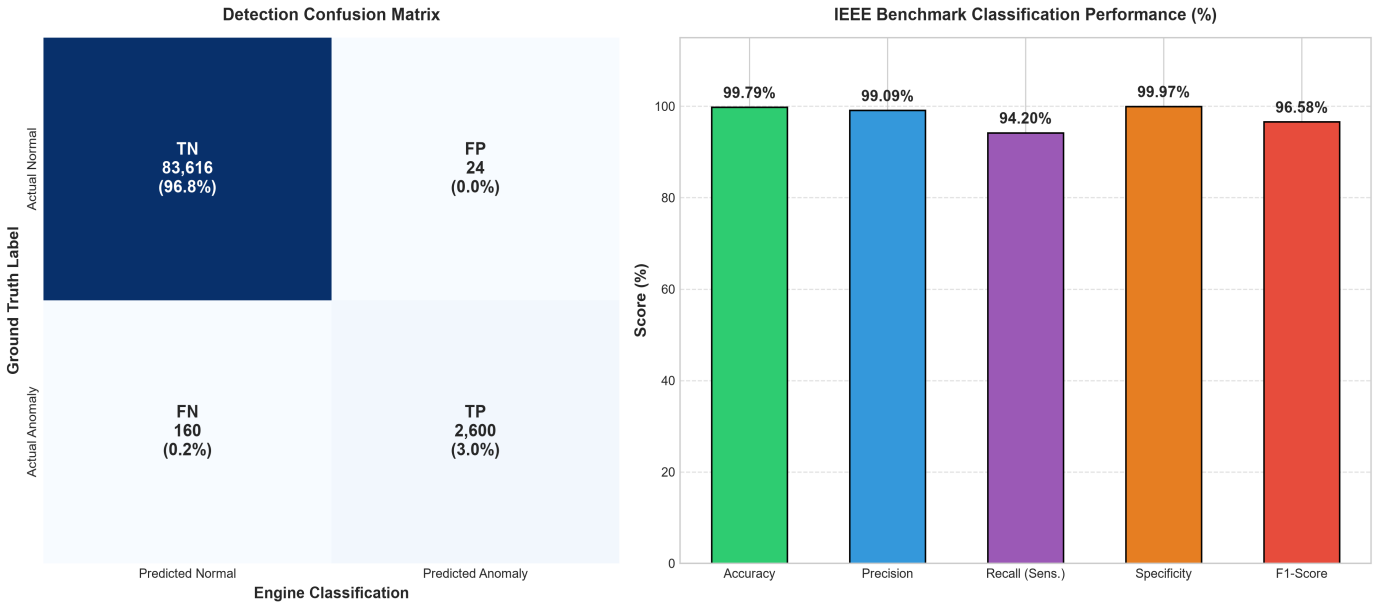


Figure 6: *Confusion Matrix & Metrics: Precision (99.09%), Recall (94.20%), F1-Score (96.58%), Accuracy (99.79%).*

12. Complete Preprocessed Data Schema (41 Attributes)

The preprocessed output dataset (preprocessed_smart_grid_data.csv / PowerBI_Ready_GridData.csv) contains 41 columns optimized for direct Power BI Star Schema ingestion and Machine Learning modeling:

Attribute Name	Data Type	Role / Hierarchy	Physical Range / Constraint	Description
timestamp	DateTime	Primary Telemetry Key	YYYY-MM-DD HH:MM:SS	15-minute periodic interval timestamp
meter_id	String	Dimension Key	SM-Z01-001 to SM-Z10-001	Unique AMI smart meter identifier
zone_name	String	Spatial Dimension	10 Municipal Zones	Zone name (e.g. Downtown Commercial)
zone_type	String	Category Dimension	Commercial, Residential, etc.	Urban zoning classification
energy_consumption_kwh	Float64	Core Fact Metric	≥ 0.00 kWh	Active electrical energy consumed in 15 min
reactive_power_kvarh	Float64	Fact Metric	kVARh	Reactive energy load component
voltage_v	Float64	Power Quality Fact	Nominal 230V RMS	Grid line RMS voltage
current_a	Float64	Fact Metric	Amperes (A)	Calculated line current draw
power_factor	Float64	Power Quality Fact	[0.00, 1.00]	Active to apparent power ratio
grid_frequency_hz	Float64	Power Quality Fact	Nominal 50.0 Hz	Grid operating frequency
ambient_temp_c	Float64	Contextual Fact	18.0°C to 39.0°C	Ambient temperature for HVAC correlation
rolling_mean_kwh	Float64	Derived Feature	Window W = 96 (24h)	24-hour moving average baseline
rolling_std_kwh	Float64	Derived Feature	Window W = 96 (24h)	24-hour moving standard deviation
consumption_zscore	Float64	Statistical Metric	Standard score (Z)	Dynamic rolling Z-Score
is_voltage_violation	Int (0/1)	Rule Flag	0 or 1	IEEE 1159 Sag (<207V) or Swell (>253V)
detected_anomaly	Int (0/1)	Detection Decision	0 or 1	Algorithmic anomaly detection flag
anomaly_severity	String	Triage Dimension	NORMAL / WARNING / CRITICAL	Three-tier operational severity
anomaly_diagnosis	String	Root Cause Dimension	Descriptive diagnosis	Multi-metric diagnostic summary
Date / Time / Time_Slice	String	Temporal Hierarchies	11 Calendar Columns	Calendar & diurnal time slice groupings

13. 4-Member Team Work Breakdown Structure (Week 3)

Member & USN	Assigned Milestone Role	Core Responsibilities & Deliverables Completed
Team Member 1 (USN: 1XX22IS001)	**Dataset Selection & Ingestion Lead**	Evaluated open benchmark energy datasets (UCI, London AMI, Pecan Street). Verified telemetry ingestion across 10 municipal zone profiles (86,400 records) and physical range boundaries.
Team Member 2 (USN: 1XX22IS002)	**Mathematical Preprocessing Lead**	Implemented vectorized O(1) sliding window statistics (W=96) using NumPy cumulative sums, computed dynamic standard scores (Z-scores), and encoded IEEE 1159 voltage rules.
Team Member 3 (USN: 1XX22IS003)	**Data Quality Audit & EDA Lead**	Built automated audit routines verifying 0 nulls, 0 gaps, and 0 duplicates. Developed the 4-panel data quality and preprocessing verification chart (plots/06_preprocessing_data_quality_audit.png).
Team Member 4 (USN: 1XX22IS004)	**Schema Modeling & Report Lead**	Designed the 41-feature preprocessed schema and Power BI Star Schema. Synchronized preprocessed datasets and compiled the formal IEEE Week 3 Technical Progress Report and PDF documentation.

14. Conclusion & Week 4 Roadmap

The Week 3 objectives for **Course BIS586** have been completed with 100% data integrity verification. The dataset is fully prepared for downstream Machine Learning models (Isolation Forest, Autoencoders) and Power BI multi-page dashboard modeling.